

Project Charter

Client

Regional Hospital System

A multi-hospital healthcare organization seeking to improve quality of care, reduce avoidable readmissions, and strengthen operational performance.

Engagement

Hospital Readmission Reduction Analytics

An end-to-end healthcare analytics consulting case study designed to identify factors associated with 30-day hospital readmissions and recommend data-driven interventions that improve patient outcomes and reduce unnecessary healthcare utilization.

Project Background

Hospital leadership has observed an increase in 30-day readmissions across several service lines. Elevated readmission rates negatively affect patient outcomes, increase operational burden, and may result in reduced reimbursement under the Centers for Medicare & Medicaid Services (CMS) Hospital Readmissions Reduction Program (HRRP).

Leadership has commissioned the Healthcare Analytics Team to investigate the underlying drivers of readmissions and identify opportunities for targeted intervention.

Business Goal

Reduce avoidable 30-day hospital readmissions through evidence-based clinical and operational improvements supported by healthcare analytics.

Primary Stakeholders

Stakeholder	Primary Interest
Chief Executive Officer (CEO)	Financial sustainability, quality performance, strategic oversight
Chief Quality Officer (CQO)	Readmission reduction, quality improvement, patient outcomes
Chief Medical Officer (CMO)	Clinical quality, evidence-based care, physician engagement
Director of Care Management	Care coordination, discharge planning, transitional care
Hospital Operations Leadership	Resource utilization, bed capacity, operational efficiency

Consulting Team

Healthcare Analytics Consulting Team

Project Leader: Wai Fun Leong

Role: Provide data-driven insights that support executive decision-making through data integration, statistical analysis, predictive modeling, visualization, and actionable recommendations.

Scope of Work

This engagement includes:

- Business problem definition
- Data quality assessment
- SQL-based ETL and data integration
- Exploratory data analysis
- Statistical analysis
- Predictive modeling for readmission risk

- Executive dashboard development
- Executive recommendations
- Final reporting and presentation

Out of Scope

The following activities are outside the scope of this engagement:

- Primary data collection (e.g., patient surveys, clinical observations, or prospective data collection).
 - Electronic Health Record (EHR) system implementation or integration.
 - Deployment of a production-ready clinical decision support system or mobile application.
 - Real-time patient risk prediction in a live clinical environment.
 - Clinical validation or implementation of predictive models for patient care.
 - Regulatory approval, compliance auditing, or changes to hospital policies and procedures.
 - Post-implementation monitoring and measurement of intervention outcomes.
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Assumptions

This engagement assumes that:

- The provided datasets accurately represent hospital operations and patient encounters.
 - Patient data have been de-identified to protect privacy.
 - Historical data are sufficient to evaluate readmission patterns.
 - Stakeholders have agreed on the definition of a 30-day readmission.
 - The analytical findings are intended to support decision-making and should complement, not replace, clinical judgment.
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Expected Deliverables

- Executive summary
- Data quality assessment report
- SQL ETL workflow

- Exploratory data analysis
 - Predictive readmission model
 - Tableau executive dashboard
 - Business recommendations
 - GitHub documentation
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Success Criteria

The engagement will be considered successful if it:

- Identifies key demographic, clinical, and operational factors associated with 30-day readmissions.
 - Provides clear, data-driven insights that support executive and clinical decision-making.
 - Develops a predictive risk stratification model capable of identifying patients at elevated risk for readmission.
 - Delivers an interactive executive dashboard for monitoring readmission trends and key performance indicators.
 - Presents practical, evidence-based recommendations to support quality improvement and reduce avoidable readmissions.
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Project Timeline

Phase	Activity
Week 1	Business discovery & project planning
Week 2	Data audit & SQL ETL
Week 3	Exploratory data analysis
Week 4	Statistical analysis & predictive modeling
Week 5	Dashboard development
Week 6	Executive recommendations & presentation